

GEOG 247/347 Assignment 8

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Part 1

```
#install.packages("gstat")  
#install.packages("viridis")
```

```
library(sp)  
library(sf)
```

Warning: package 'sf' was built under R version 4.5.3

Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE

```
library(gstat)
```

Warning: package 'gstat' was built under R version 4.5.3

```
library(ggplot2)  
library(tmap)
```

Warning: package 'tmap' was built under R version 4.5.3

```
# Load the meuse sample data  
data(meuse, package = "sp")
```

```
#?meuse
```

```
meuse_sf <- st_as_sf(meuse, coords = c("x", "y"), crs = 28992)

tm_shape(meuse_sf) +
  tm_dots(col = "zinc",
          style = "quantile",
          palette = "Reds",
          size = 0.3) +
  tm_scalebar(position = c("right", "bottom")) +
  tm_layout(title = "Zinc Concentration in Meuse",
            frame = FALSE)
```

-- tmap v3 code detected -----

[v3->v4] `tm\_dots()`: instead of `style = "quantile"`, use `fill.scale =`
`tm_scale_intervals()`.

i Migrate the argument(s) 'style', 'palette' (rename to 'values') to
`'tm_scale_intervals(<HERE>)'`

[v3->v4] `tm\_dots()`: use 'fill' for the fill color of polygons/symbols
(instead of 'col'), and 'col' for the outlines (instead of 'border.col').

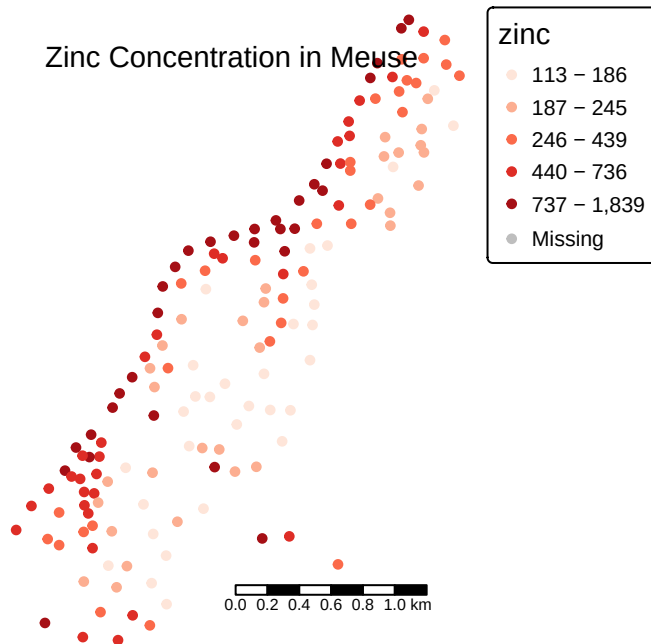
[v3->v4] `tm\_layout()`: use `tm_title()` instead of `tm_layout(title = )`

[cols4all] color palettes: use palettes from the R package cols4all. Run
`'cols4all::c4a_gui()'` to explore them. The old palette name "Reds" is named
"brewer.reds"

Multiple palettes called "reds" found: "brewer.reds", "matplotlib.reds". The first one, "brewer.reds"

[plot mode] fit legend/component: Some legend items or map components do not
fit well, and are therefore rescaled.

i Set the tmap option `'component.autoscale = FALSE'` to disable rescaling.

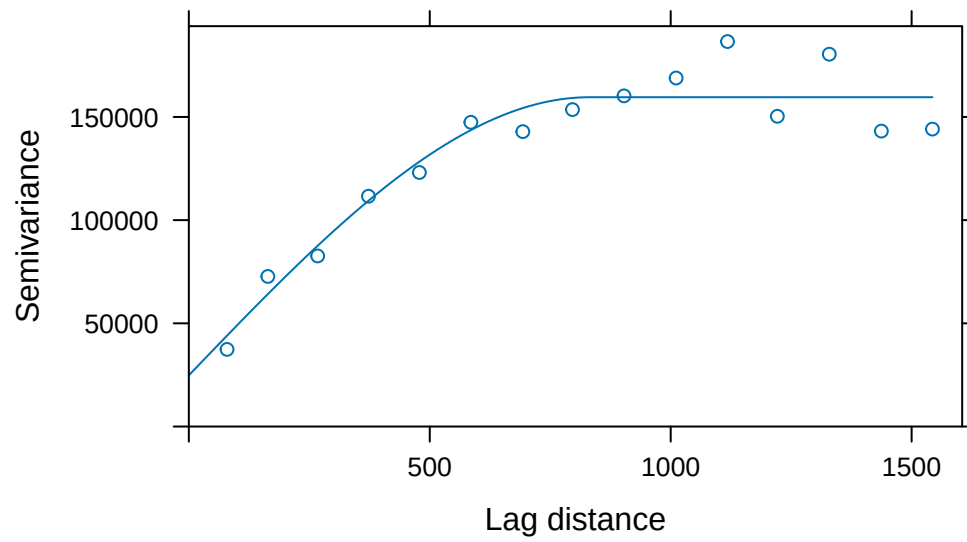


```
vgm_emp <- variogram(zinc ~ 1, data = meuse_sf)

# fit spherical model
vgm_fit <- fit.variogram(vgm_emp, vgm("Sph"))

plot(vgm_emp, vgm_fit,
     main = "Empirical Semivariogram of Zinc",
     xlab = "Lag distance",
     ylab = "Semivariance",
     pch = 1)
```

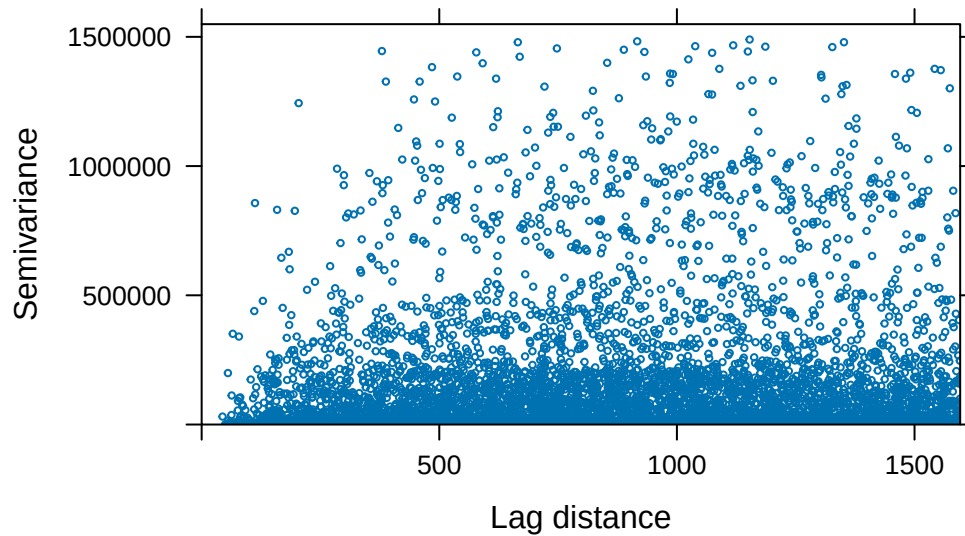
Empirical Semivariogram of Zinc



```
vgm_cloud <- variogram(zinc ~ 1, meuse_sf, cloud=TRUE)
```

```
plot(vgm_cloud,  
     main = "Variogram cloud of Zinc",  
     xlab = "Lag distance",  
     ylab = "Semivariance",  
     pch = 1,  
     cex = 0.4)
```

Variogram cloud of Zinc



```
vgm_emp <- variogram(zinc ~ 1, meuse_sf)

vgm_sph <- fit.variogram(vgm_emp, vgm("Sph"))
vgm_exp <- fit.variogram(vgm_emp, vgm("Exp"))
vgm_gau <- fit.variogram(vgm_emp, vgm("Gau"))

line_sph <- variogramLine(vgm_sph, maxdist = max(vgm_emp$dist))
line_exp <- variogramLine(vgm_exp, maxdist = max(vgm_emp$dist))
line_gau <- variogramLine(vgm_gau, maxdist = max(vgm_emp$dist))

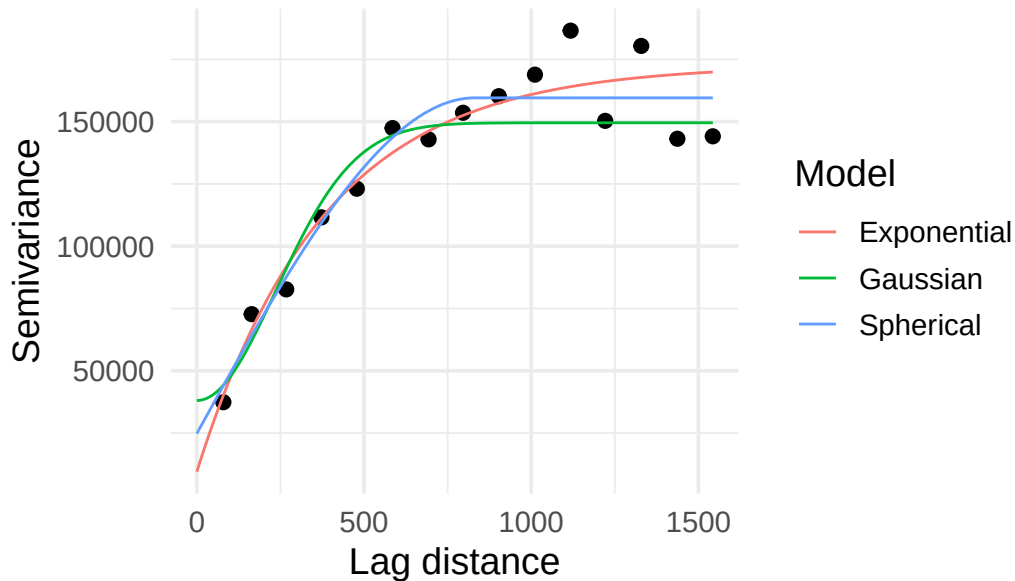
line_sph$model <- "Spherical"
line_exp$model <- "Exponential"
line_gau$model <- "Gaussian"

lines_df <- rbind(line_sph, line_exp, line_gau)

ggplot() +
  geom_point(data = vgm_emp, aes(x = dist, y = gamma), size = 2) +
  geom_line(data = lines_df, aes(x = dist, y = gamma, color = model), linewidth = 0.5) +
  labs(
    title = "Empirical Semivariogram with Fitted Models",
    x = "Lag distance",
    y = "Semivariance",
```

```
color = "Model"
) +
theme_minimal(base_size = 14)
```

Empirical Semivariogram with Fitted Models



```
attr(vgm_sph, "SSErr")
```

```
[1] 2223257
```

```
attr(vgm_exp, "SSErr")
```

```
[1] 1791466
```

```
attr(vgm_gau, "SSErr")
```

```
[1] 3729204
```

Task a

When lag distance is relatively small, the zinc sample-based semivariance rises sharply. As lag distance increases, especially past the range value (around 750), semivariance stabilizes around 150,000 but the data becomes noisier.

Task b

I think the exponential variogram model fits this empirical semivariogram the best. The exponential model has the lowest SSE value of 1791466 which indicates that the model more closely follows the empirical semivariogram. Also, visually speaking, the exponential model appears to more closely align with the zinc samples semivariance values as lag distance increases.

Task c

In the context of this semivariogram, the range (extending from 0 to ~750) represents the lag distance beyond which zinc sample values similarity is no longer spatially dependent. That is to say, the range tells me that after ~750 meters the zinc concentration values are no longer spatial autocorrelated.

Part 2

```
library(sf)
library(gstat)
library(tmap)
library(viridis)
```

Warning: package 'viridis' was built under R version 4.5.3

Loading required package: viridisLite

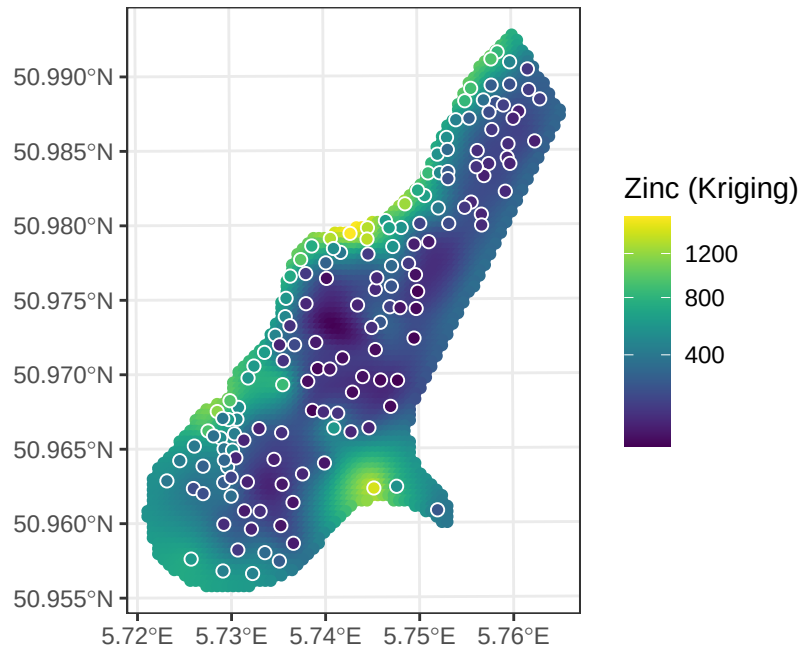
```
library(viridisLite) #loading the other libraries without this one gave me a warning to also
```

```
data(meuse.grid)
meuse.grid <- st_as_sf(meuse.grid, coords = c("x", "y"),
                      crs = 28992)
```

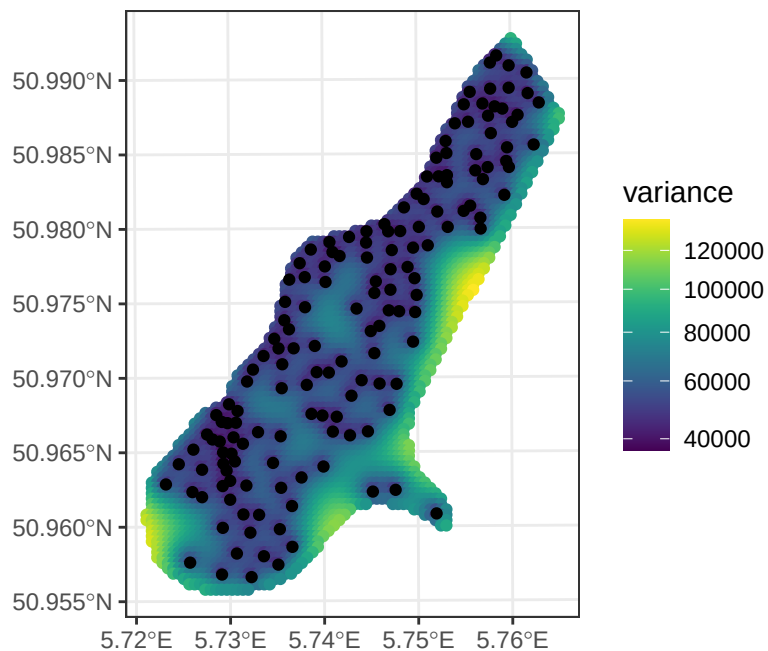
```
kpred <- krige(zinc ~ 1, meuse_sf, meuse.grid, model = vgm_sph)
```

[using ordinary kriging]

```
ggplot() + geom_sf(data = kpred, aes(color = var1.pred)) +
  geom_sf(data = meuse_sf, aes(fill = zinc), shape = 21, color = "white", size = 2) +
  scale_color_viridis(name = "Zinc (Kriging)", trans = "sqrt") +
  scale_fill_viridis(guide = "none", trans = "sqrt") +
  theme_bw()
```



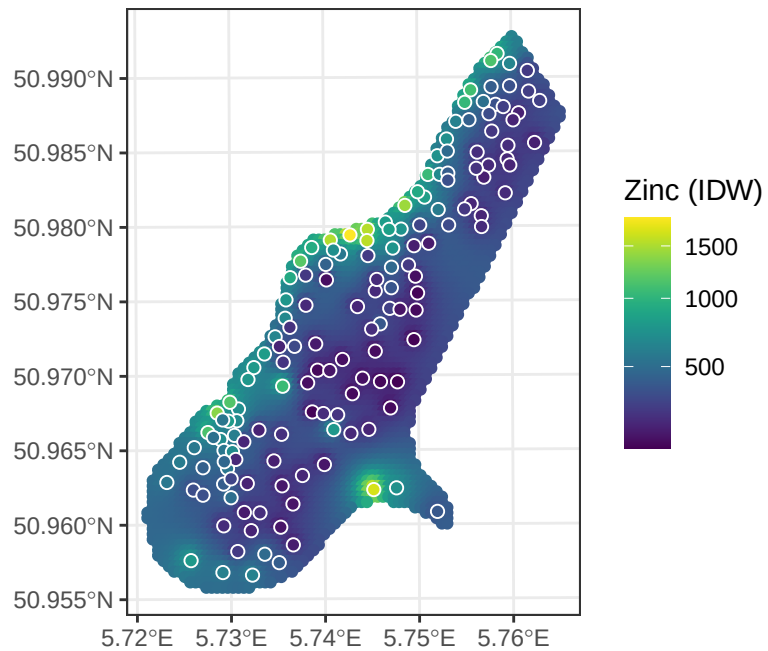
```
ggplot() + geom_sf(data = kpred, aes(color = var1.var)) +
  geom_sf(data = meuse_sf) +
  scale_color_viridis(name = "variance", trans = "sqrt") + theme_bw()
```

```
idw_pred <- idw(zinc ~ 1, meuse_sf, meuse.grid, idp = 2)
```

[inverse distance weighted interpolation]

```
ggplot() +
  geom_sf(data = idw_pred, aes(color = var1.pred)) +
  geom_sf(data = meuse_sf, aes(fill = zinc), shape = 21, color = "white", size = 2) +
  scale_color_viridis(name = "Zinc (IDW)", trans = "sqrt") +
  scale_fill_viridis(guide = "none", trans = "sqrt") +
  theme_bw()
```

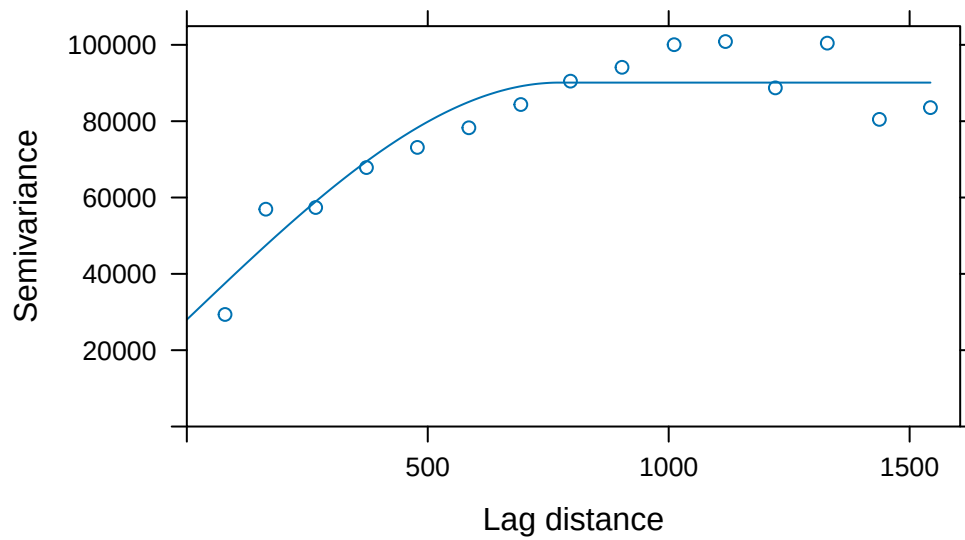


```
# empirical variogram after accounting for distance to river
vgm_emp_dist <- variogram(zinc ~ dist, data = meuse_sf)

# fit spherical model
vgm_fit_dist <- fit.variogram(vgm_emp_dist, vgm("Sph"))

# plot
plot(vgm_emp_dist, vgm_fit_dist,
     main = "Variogram for Universal Kriging",
     xlab = "Lag distance",
     ylab = "Semivariance",
     pch = 1)
```

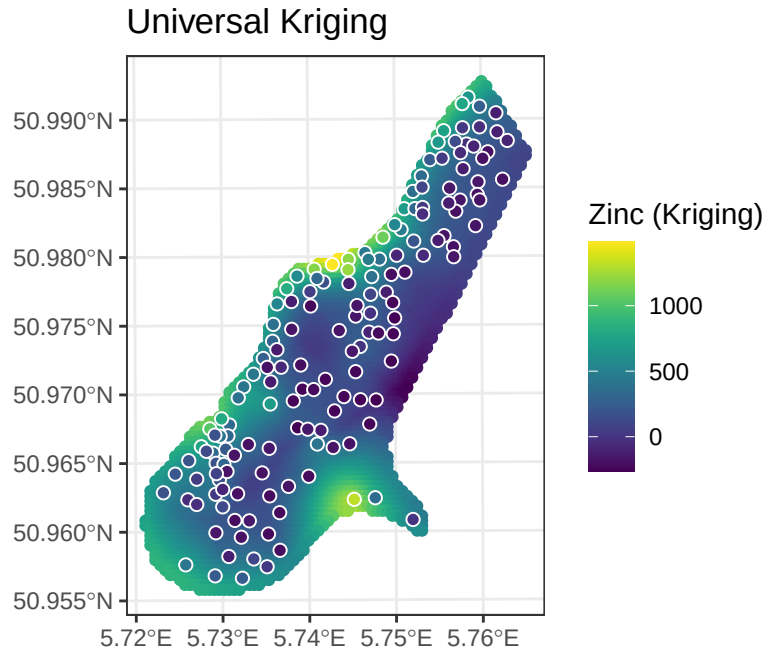
Variogram for Universal Kriging



```
kpred_dist <- krige(zinc ~ dist, meuse_sf, meuse.grid, model = vgm_fit_dist)
```

[using universal kriging]

```
ggplot() + geom_sf(data = kpred_dist, aes(color = var1.pred)) +  
  geom_sf(data = meuse_sf, aes(fill = zinc), shape = 21, color = "white", size = 2) +  
  scale_color_viridis(name = "Zinc (Kriging)") +  
  scale_fill_viridis(guide = "none") +  
  ggtitle("Universal Kriging") +  
  theme_bw()
```



Task a

The universal Kriging plot map above show a higher concentrations along the south and southwest edges of the dataset as well as along the northern edge that spans from southwest to north east. The lowest concentration is along the middle eastern edge of the dataset with a pattern of low concentration sample values throughout the middle (Center) of the study area.

Task b

The kriging variance is relatively high in the center of the eastern edge of the study area, along the edge toward the south eastern edge, and on the south western edge as well. I think this is caused by the lack of sample locations (points) around these high variance areas. The farther an area is from a sample point location, the higher the variance value is because there isn't enough information for the estimate which diminishes the models ability to interpolate values accurately.

Task c

Due to IDWs simplistic and deterministic interpolation method which weights point values based on inverse distance, sample areas that are close together have higher weights and sample areas that are farther away from each other have lower weights. Kriging uses variogram results to assign weighting to point values which account for lag distance and semivariance

values (a sample points spatial dependence) for each sample area. The kriging, due to the complexity of the variogram weighting method and inclusion of the variance values, shows the likelihood of interpolated values based on prediction variance. This results in the kriging being a more smooth surface with more than inverse distance factoring into weighting, while the IDW produces less smooth surfaces with a more prominent “bullseye” effect. The “bullseye” effect is most clear in the high zinc sample point in the southeastern edge of the study area. In the kriging map, this weighted value creates a wider gradient of higher interpolated values around the point, while the IDW map shows this point as a high value observation with the surrounding interpolated values quickly dropping off in value - creating a more stark contrast in interpolation. Similarly but with low value sample points, kriging map shows the interpolated values toward the center of the study area as lower based on the surrounding values and variogram weighting factors. This same location mapped using the IDW method doesn’t pick up on this trend of lower interpolation values and shows more uniform zinc concentration values due to the simplistic inverse distance weighting method.